Analysis of the Effects of Climatic Variables on Snow Cover Variability in Northern Pakistan

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Abstract

- Northern Pakistan's water security is increasingly threatened by diminishing snow cover, exacerbated by climate change.
- Two-decade analysis (2003-2023) utilizing MODIS and GLDAS-Model data reveals snow dynamics' intricacies.
- Strong negative correlation between Snow Cover Area and temperature underscores climate's influence.
- Insights gained are instrumental in formulating adaptive water management strategies for the region.
- Urgent interventions are imperative to mitigate the impact of declining snow resources.
- This study offers actionable insights to bolster water security amid evolving climate conditions in Northern Pakistan.
- Collaborative efforts are needed to implement sustainable solutions and safeguard the region's water future.
- Continued monitoring and research are crucial for adapting to changing environmental conditions and ensuring long-term water sustainability.

Introduction

- Snow cover, integral to the global water cycle and climate regulation, is significantly influenced by interactions with air temperature.
- Research spanning 2003 to 2023 in Northern Pakistan examines Snow Cover Area, Snow Melt, Air Temperature, and Land Surface Temperature (LST), utilizing remote sensing technology for precise mapping.
- Glaciers, covering 10% of Earth's surface, play a crucial role in river runoff and freshwater supply.
- Northern Pakistan's heavy reliance on snowmelt for dam reservoirs underscores the urgency of addressing climate-induced snow cover depletion.
- The region's diverse topography and climatic intricacies, coupled with its sensitivity to climate change, necessitate adaptive strategies.
- This research aims to discover the complex relationships between climate dynamics and snow cover, guiding sustainable environmental management and water resource planning.
- Understanding climate change implications on snow cover dynamics is pivotal for effective adaptation and mitigation strategies.
- Collaborative efforts involving scientists, policymakers, and local communities are essential for implementing sustainable solutions and safeguarding water resources in Northern Pakistan.

Study Area

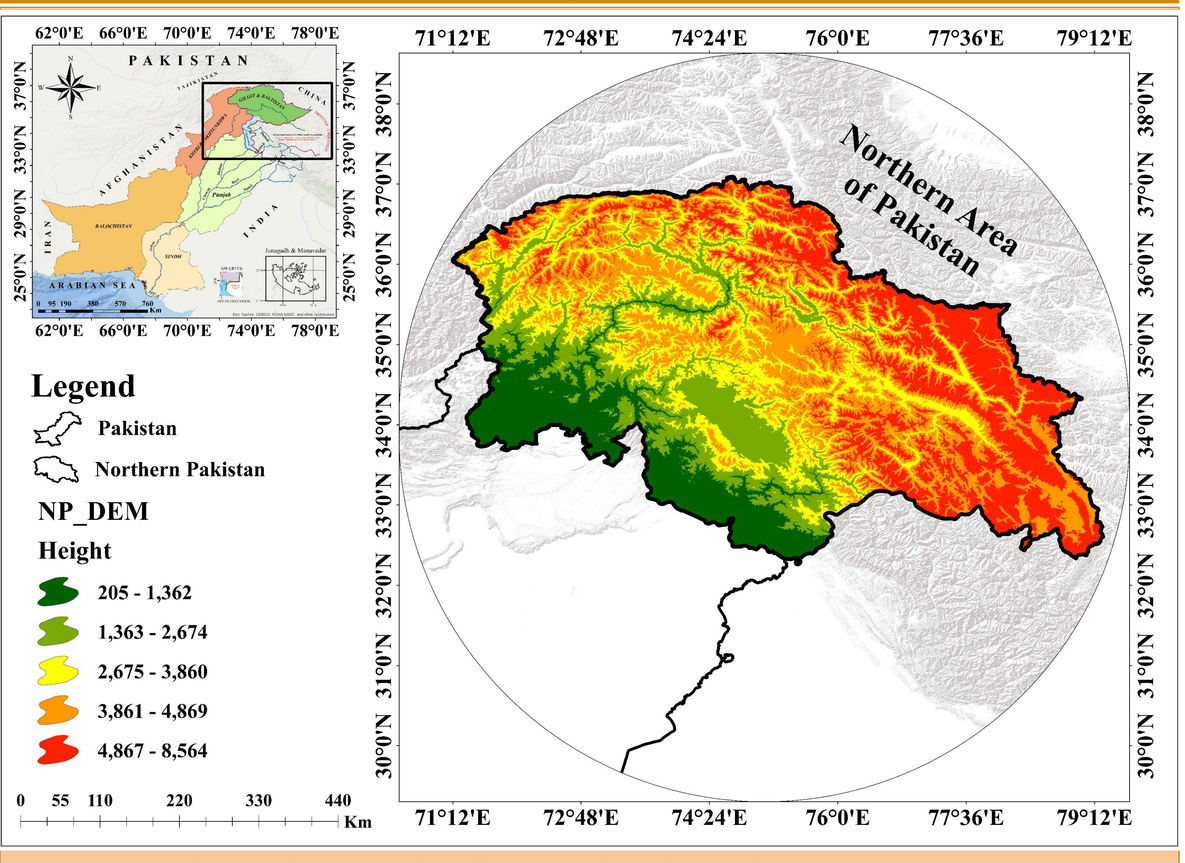


***Figure**: Map showing the geographical location of Northern Pakistan (36.5°N, 73.5°E).*

Materials and Methods

MODIS-Terra:
On NASA's Terra satellite, launched in 1999, monitors air pollution, radiation, and clouds with a temporal resolution of one to two days and swath width of 2330 km, operating across thirty-six spectral bands from 0.4 µm to 14.4 µm.

GLDAS-Model:
Integrates satellite and ground-based measure-ments for global modeling, providing 3-hourly data from the GLDAS Noah model at spatial resolution of 0.25 degrees for analysis of Air Temperature, Snow Melt, and real-time predictions.

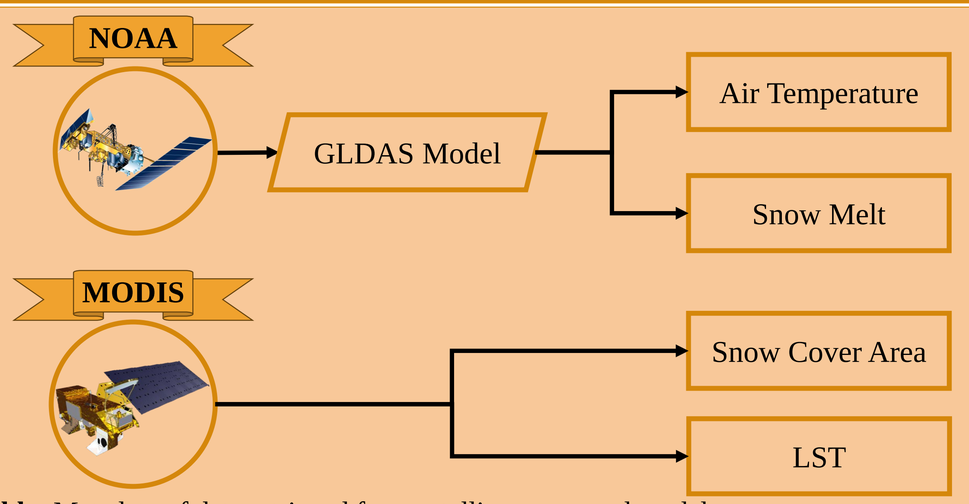
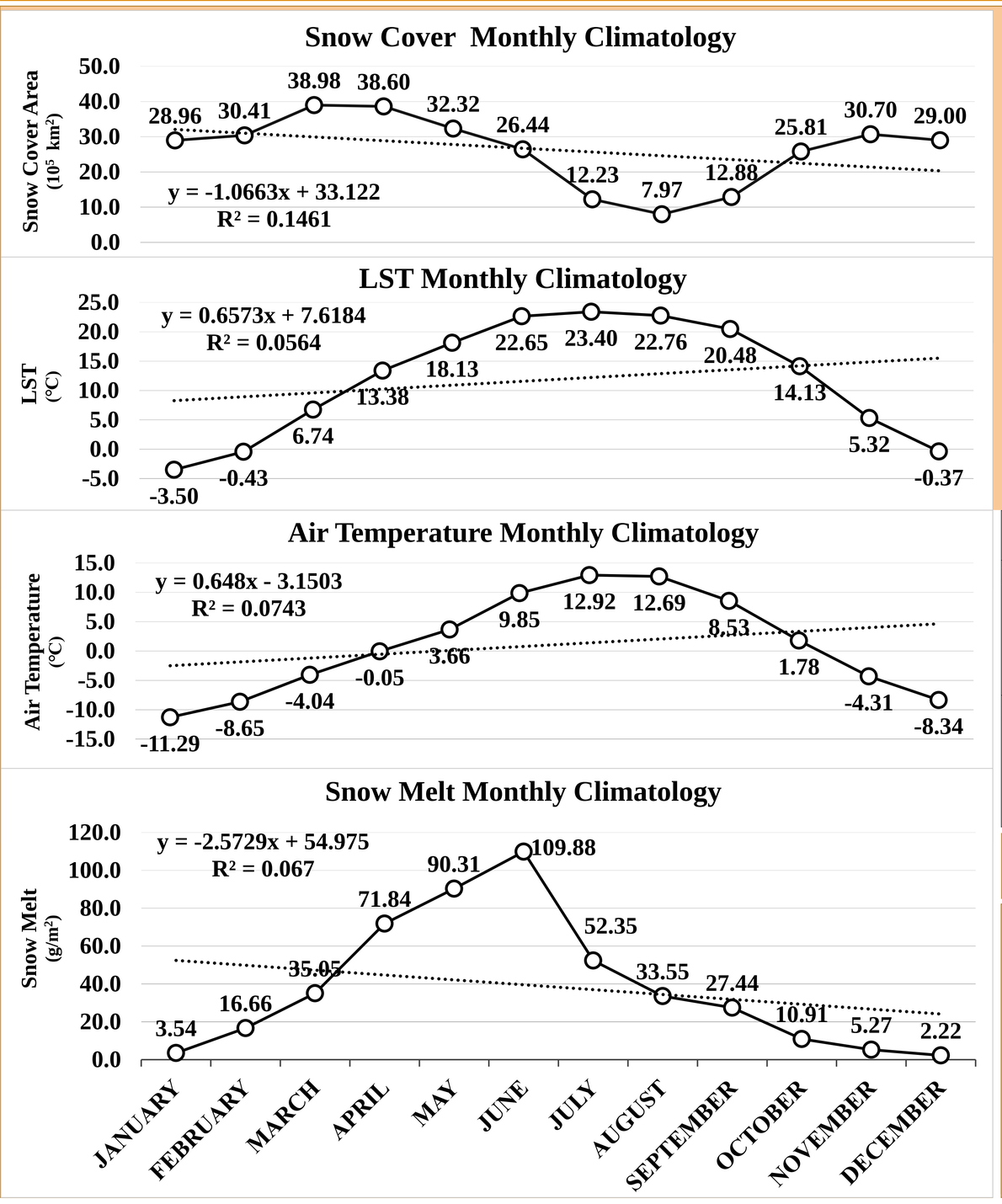


Table: Metadata of data retrieved from satellite sensor and models.

PRODUCT NAME/ID	SOURCE	UNITS	SPATIAL RESOLUTION	TEMPORAL RESOLUTION
Snow Melt – GLDAS_NOAH025_3H V2.0	GLDAS	kg/m^2	0.25°	3 Hours
Air Temperature – GLDAS_NOAH025_3H V2.1	GLDAS	Kelvin (K), Celsius (^{O}C)	0.25°	3 Hours
Snow Cover Area – MODIS/061/MOD10A1	MODIS	km^2	500 m	1 Day
LST – MOD11A1.061	MODIS	Kelvin (K), Celsius (^{O}C)	1 Km	1 Day

Results and Discussion



Monthly Climatology Analysis:

- Snow Cover Area peaks in winter, especially in January and February (38.98 × 10^5 km^2).
- Snow Melt peaks in spring, with variations across years.
- Air Temperature varies seasonally, with colder winters and warmer summers.
- Land Surface Temperature (LST) is highest in summer (23.4 ℃).
- These insights illuminate the interaction between snow dynamics and temperature changes.

Correlation Analysis:

- Positive correlations between Snow Melt and Air Temperature ($r = 0.46$, $p<0.01$), and Snow Melt and LST ($r = 0.54$, $p<0.01$), indicate temperature's influence on snow dynamics.
- Strong negative correlations between Snow Cover Area and both Air Temperature ($r = -0.62$, $p<0.01$) and LST ($r = -0.52$, $p<0.01$) highlight temperature's impact on snow coverage.
- Strong positive correlation between LST and Air Temperature ($r = 0.97$, $p<0.01$) underscores their relationship.
- Statistical significance ($p<0.01$) reinforces the reliability of these associations, offering valuable insights into environmental dynamics.

Pearson Correlation and Statistical Significance				
Pearson Correlation	**Parameter**	**Snow Cover Area**	**Snow Melt**	**Air Temperature**
	Snow Melt	0.148		
	Air Temperature	***-0.622***	***0.459***	
	LST	***-0.515***	***0.535***	***0.974***
Sig. (1-tailed)	**Parameter**	**Snow Cover Area**	**Snow Melt**	**Air Temperature**
	Snow Melt	0.130		
	Air Temperature	***0.000***	***0.000***	
	LST	***0.000***	***0.000***	***0.000***

Conclusion

- Conducted comprehensive analysis of Snow Cover Area, Snow Melt, Air Temperature, and LST in Northern Pakistan using MODIS and GLDAS Model data.
- Empirically proved distinct seasonal patterns and highlighted temperature's influence on snow dynamics, revealing positive correlations between Snow Melt and Air Temperature/LST.
- Negative correlation between Snow Cover Area and Air Temperature provided further insights into environmental dynamics.
- Study contributes valuable knowledge on region's sensitivity to climate variations, informing decision-making in water resource management and climate resilience.